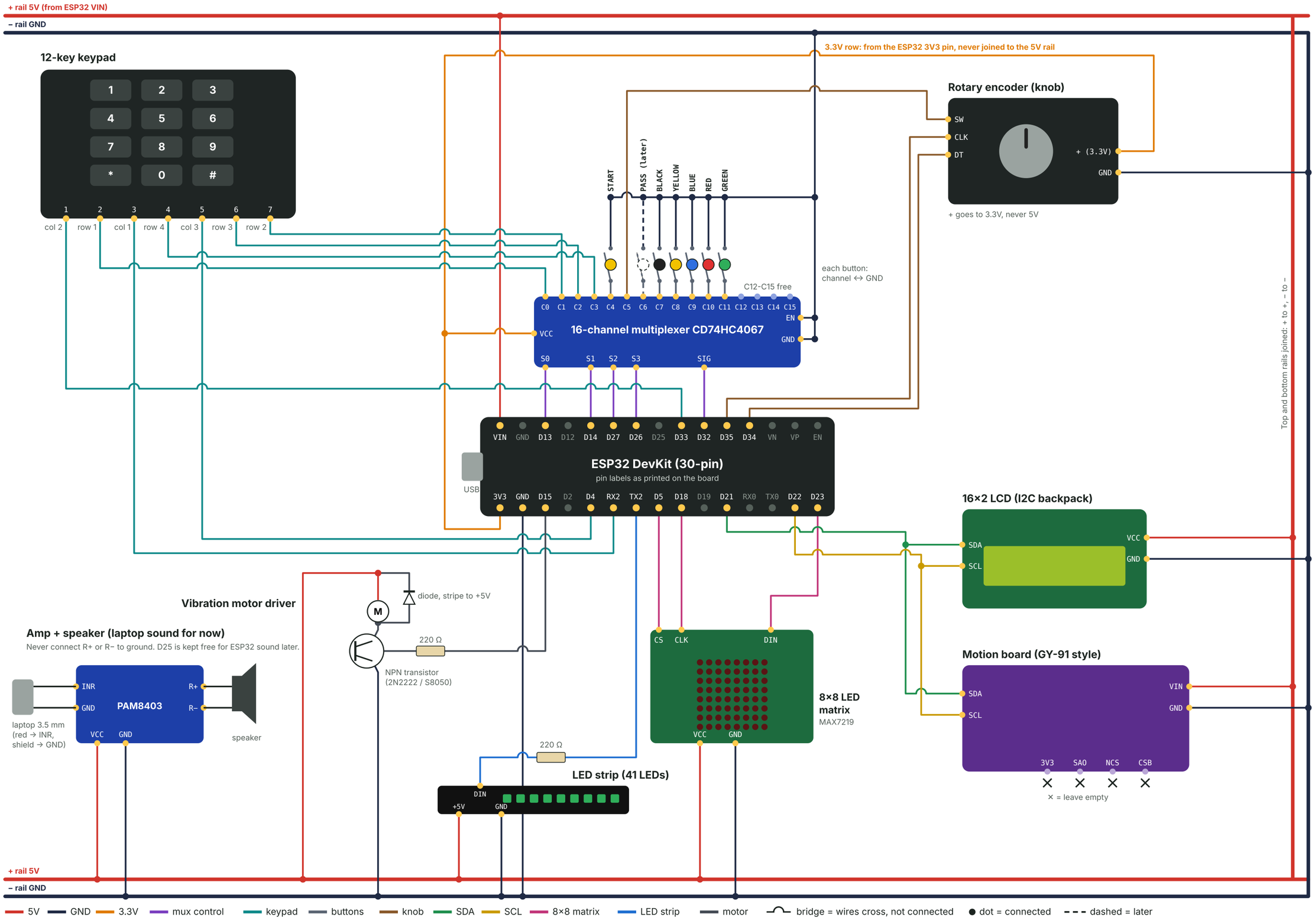
Reset Protocol: whole wiring (multiplexer build)

Every wire from the ESP32 to every part. Colours show what each wire does. Updated 20 Oct 2026.

Not built yet: the bench is still on the original wiring.
After the rewire, set **UseMux = true** in the sketch.

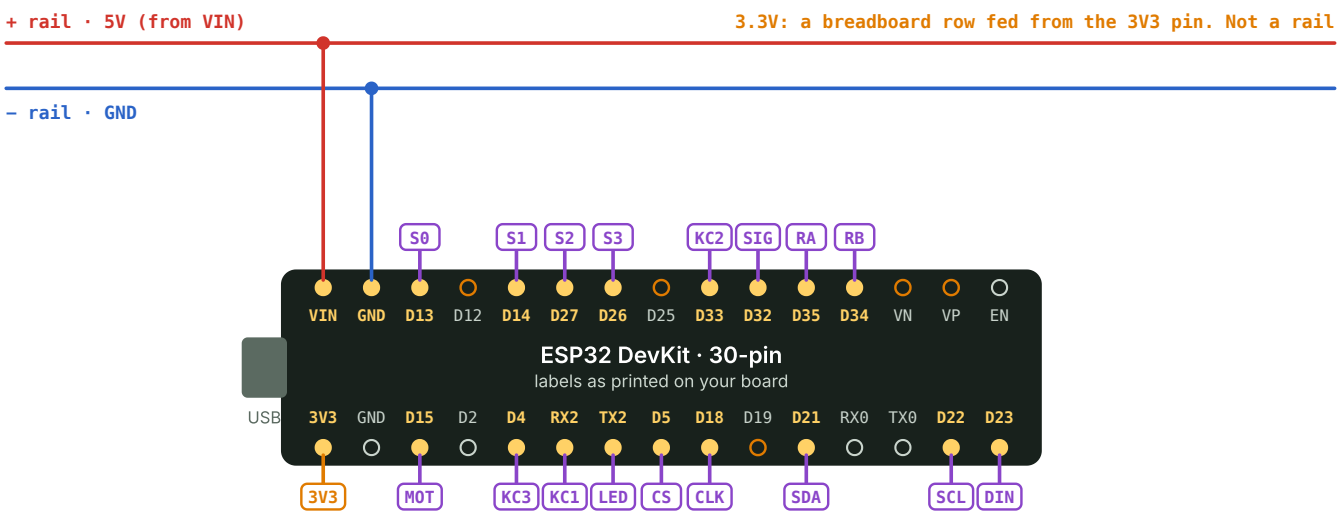


BENCH WIRING V2 · WITH THE 16-CHANNEL MULTIPLEXER · UPDATED 20 OCT 2026

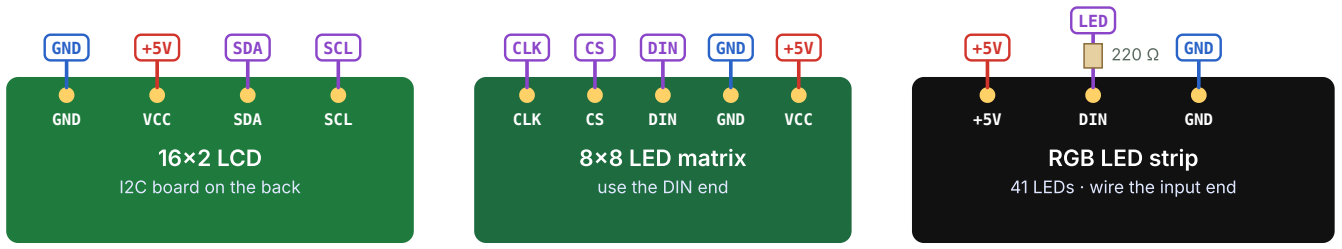
Bomb Wiring v2

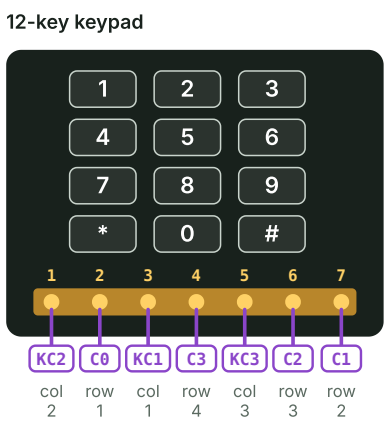
The multiplexer reads the keypad rows, the start button, the knob press and the five colour buttons on the cardboard panel (black, yellow, blue, red, green). The motion board shares SDA and SCL with the LCD. D25 is kept for ESP32 sound later. Every connection is drawn as a tag: two pins with the same tag share one wire.

What changed on 20 Oct: the colour buttons are now wired for real (green added on C11), the motion board is on the drawing, and the free pins are updated. Pass on C6 is still for later.

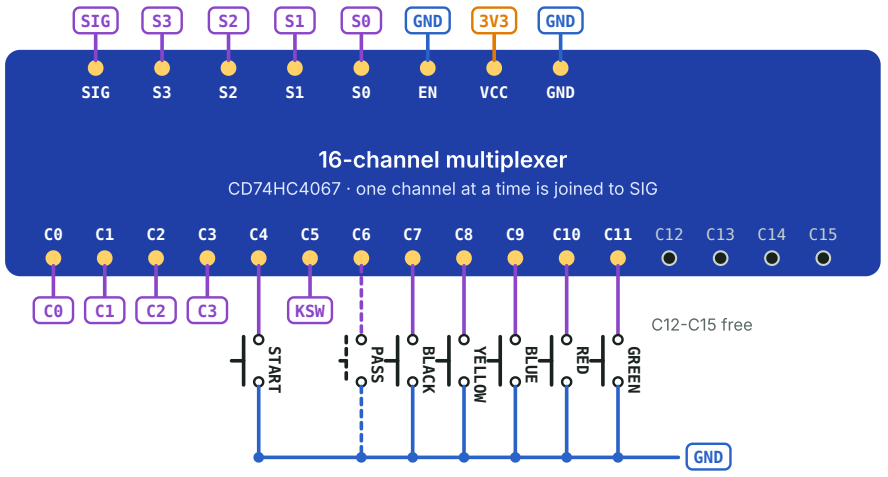


Hollow orange pins: D25 is kept for ESP32 sound (later). D19, D12, VP and VN are free. D2 drives the blue LED.



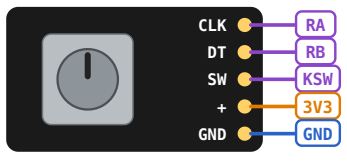


Columns (KC) go to the ESP32. Rows (C0-C3) go to the multiplexer. No power needed.



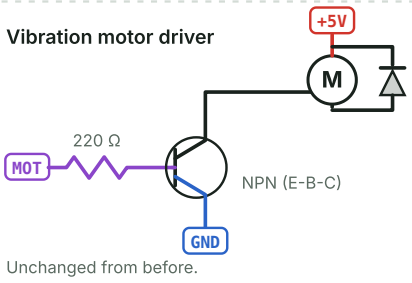
Every button: one leg to its channel, the other to the − rail.
The 5 colour buttons sit on the cardboard panel. Pass (dashed) is for later.

Rotary encoder (knob)



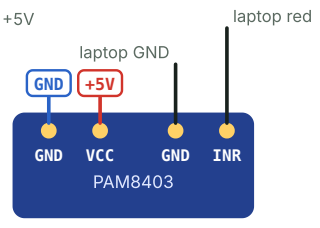
+ goes to 3.3V, never 5V.

Vibration motor driver



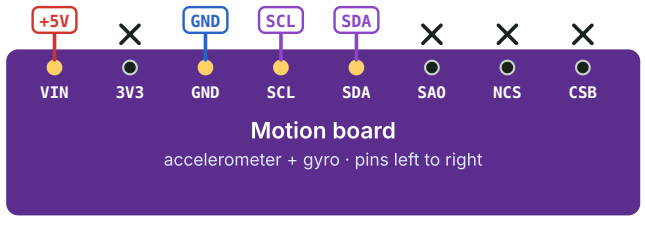
Unchanged from before.

Amp + speaker (laptop sound)



Speaker on R+ / R−. D25 is kept for when the ESP32 plays sound itself.

Motion board (GY-91 style) · shares SDA / SCL with the LCD



SCL and SDA go in the same breadboard rows as the LCD's SCL and SDA.
One numbered row each, with an empty row between them.
× = leave empty. The board makes its own 3.3V.
Boot log should say: Motion chip at 0x68.

5V · 3.3V · Ground · Signal · Dashed = Pass button, for later

Rewire from v1, step by step

1. **Unplug USB.**
2. **Remove** the test LED and its resistor on D33. The ESP32's own blue LED takes over that job.
3. **Keypad.** Pull all 7 keypad wires off the ESP32, then plug them back like this: `1 → D33 · 2 → mux C0 · 3 → RX2 · 4 → mux C3 · 5 → D4 · 6 → mux C2 · 7 → mux C1`
4. **Start button:** move the wire from RX2 (D16) to **mux C4**. Its other leg stays on the – rail.
5. **Encoder SW:** move the wire from VP to **mux C5**.
6. **3.3V row:** jumper from the ESP32's **3V3** pin to an empty breadboard row. Move the encoder's **+** into that row.
7. **Mux power:** VCC to the 3.3V row. GND and EN to the – rail.
8. **Mux control:** S0 → D13, S1 → D14, S2 → D27, S3 → D26, SIG → D32.
9. **Colour buttons** (cardboard panel): black → **C7**, yellow → **C8**, blue → **C9**, red → **C10**, green → **C11**. Each button's other leg to the – rail. You can chain those ground legs together on the panel and run one wire to the rail.
10. **Motion board:** leave it as it is: VIN on +, GND on –, SCL and SDA in the LCD's rows.
11. Plug USB back in, set `UseMux = true` in the sketch and upload. In the Serial Monitor, press each keypad key, the start button, the knob and every colour button to check them.

If something's off

- > 3.3V goes to its own breadboard row. Never join it to the 5V + rail.
- **No key works:** check mux EN is on ground and VCC on the 3.3V row.
- **A whole row of keys is dead:** that row's wire to C0-C3 is loose.
- **A whole column is dead:** check its wire to D33, RX2 or D4.
- **Keys come out wrong:** two keypad wires are swapped. Check the order 1-7 again.
- **Start button does nothing:** it needs C4 on one leg and the – rail on the other.
- Colour buttons print `Button RED pressed` and so on in the Serial Monitor, and light up on the wireless page.
- **A colour button does nothing:** check its two legs. On 4-leg buttons, use legs that are diagonally opposite.
- **A colour button reads as another colour:** its channel wires are swapped. Check C7-C11 against the list.
- **Motion board or LCD gone:** SDA and SCL are in the same breadboard row. Give each its own row.

Every connection

FROM	TO	WIRE	WHY
ESP32 POWER			
ESP32 `VIN`	+ rail	F–M	5V for the LCD, matrix, strip, amp, motor
ESP32 `GND`	– rail	F–M	Shared ground
ESP32 `3V3`	An empty breadboard row (the 3.3V row)	F–M	Feeds the mux and the encoder. Keep it off the 5V rail
MULTIPLEXER			
Mux `VCC`	3.3V row	F–M	3.3V keeps its signals safe for the ESP32
Mux `GND` and `EN`	– rail	F–M ×2	EN grounded = mux switched on
Mux `S0 S1 S2 S3`	ESP32 `D13 D14 D27 D26`	F–F ×4	Pick the channel
Mux `SIG`	ESP32 `D32`	F–F	Reads the picked channel
KEYPAD (PINS 1-7, LEFT TO RIGHT, 1 2 3 AT THE TOP)			
Keypad 1 (column 2)	ESP32 `D33`	F–F	
Keypad 2 (row 1)	Mux `C0`	F–F	
Keypad 3 (column 1)	ESP32 `RX2` (D16)	F–F	
Keypad 4 (row 4)	Mux `C3`	F–F	
Keypad 5 (column 3)	ESP32 `D4`	F–F	
Keypad 6 (row 3)	Mux `C2`	F–F	
Keypad 7 (row 2)	Mux `C1`	F–F	
BUTTONS (OTHER LEG OF EACH TO THE – RAIL)			

FROM	TO	WIRE	WHY
Start button	Mux C4	F–M	Short press starts, hold 1.5 s resets
Black colour button	Mux C7	F–M	Cardboard panel
Yellow colour button	Mux C8	F–M	Cardboard panel
Blue colour button	Mux C9	F–M	Cardboard panel
Red colour button	Mux C10	F–M	Cardboard panel
Green colour button	Mux C11	F–M	New. Cardboard panel
Pass button (later)	Mux C6	F–M	Not wired yet. Code already prints presses
ROTARY ENCODER			
Encoder CLK / DT	ESP32 D35 / D34	F–F ×2	Unchanged
Encoder SW	Mux C5	F–F	Moved: the mux gives it the pull-up it was missing
Encoder +	3.3V row	F–M	Never 5V
Encoder GND	– rail	F–M	
MOTION BOARD (ALREADY WIRED, UNCHANGED BY THE REWIRE)			
Motion VIN	+ rail	F–M	It has its own 3.3V regulator
Motion GND	– rail	F–M	
Motion SCL / SDA	Same rows as the LCD's SCL / SDA (D22 / D21)	F–M ×2	Shared I2C line. 3V3, SAO, NCS, CSB stay empty
UNCHANGED			
LCD SDA / SCL	D21 / D22	F–F ×2	VCC to + rail, GND to – rail
Matrix DIN / CS / CLK	D23 / D5 / D18	F–F ×3	VCC to + rail, GND to – rail
LED strip DIN	220 Ω, then TX2 (D17)		+5V and GND to the rails
Motor driver	D15 through 220 Ω to the transistor		See the motor page

F = female end (slides onto a pin). M = male end (plugs into the breadboard).